Investigating Tolerance for Uncertainty Through Text Mining: Enhancing Human Resource Strategy in Seed-Stage High-Tech Startups



Yushi Nakaya and Shuichi Ishida

1 Introduction

Background

Startup companies are poised to play a pivotal role in addressing both existing and emerging societal challenges, especially those known as "wicked problems," which are inherently complex and vexing (Stephen et al., 2021). Recent years have witnessed a surge in high-tech startups aiming to drive innovation with cutting-edge technology, setting themselves apart from businesses offering easily replicable solutions. The evolution of these startups, including changes in size and product status, occurs in parallel with their advancement through various funding rounds: the angel round, seed stage, series A, scale-up (series B and series C), and growth phases (series D and beyond). Notably, the seed stage emerges as particularly precarious and dynamic, marked by the business's establishment, identification of the ideal customer base, and the development of a Minimum Viable Product (MVP) (Failed, 2021; Tomczak & Brem, 2013; Zhang & Wang, 2017).

During the seed stage, the scope of individual work typically exceeds expectations found in larger corporations, a shift attributed mainly to a smaller workforce (Ycombinator, 2022). This phase also experiences significant fluctuations in organizational size, with personnel numbers often changing by a factor of 5 to 10 (Ycombinator, 2022).

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Therefore, it is important for all members to work independently and think about what is missing in the organization on their own, and to take a leadership and ownership role in each of their own tasks. Thus, recruiting and training individuals for leadership roles and integrating them into the organization's core team become crucial. Such efforts are vital for shaping the company's future internal culture (Tannenbaum, 1966).

Managers of seed-stage startups, including the author, face the challenge of ensuring new hires align with the company's culture without risking disruption. Past experiences have shown that misalignments can lead to mutual consent terminations, underlining the severe and potentially irreversible consequences of poor hiring decisions on organizational culture. Addressing these risks is, therefore, of paramount importance.

An employee's high job performance, in other words, competency, is known to be related to his or her positive personality (Hu et al., 2021). In Hu et al.(2021), competency theory is described as follows.

The competency theory (McClelland, 1973) claimed that competency was the core factor for improving job performance: the higher the competency, the better the job performance. Luna-Arocas and Morley (2015) defined competency as "a set of observable dimensions, including individual knowledge, skills, attitudes, and behaviors, as well as collective team, process, and organizational capabilities, that are linked to high performance and provide the organization with sustainable competitive advantage."

Therefore, in seed stage startups, where the abilities and characteristics of employees have a significant impact on creating an organizational culture that is fundamental to the company's performance, it is extremely important to correctly assess and hire for competencies that are directly related to job performance, especially in order to lower the risk of poor hiring decisions, as mentioned earlier. However, in seed stage startups, there are a large number of things that founders need to do, such as product development, needs assessment, and conversations with investors, in addition to recruiting and training. Based on the author's experience, it is very difficult to find out whether or not a person has a high level of competencies that will enable him or her to perform well, so it is important to set hiring criteria and conduct hiring activities at a low cost.

While existing research largely focuses on methodologies vital for the growth of seed-stage startups and frameworks for harnessing and scaling technological assets (Atzmon et al., 2022), as well as decision-making post-COVID-19 (Parthasarathy, 2022), less attention has been given to the specific conditions and capabilities these organizations need. Although the significance of individual competencies in human resources, their organizational alignment, and their contribution to mitigating startup failure risks—the "valley of death" (Barr et al., 2009; Cummings et al., 2018; Markham et al., 2010; Thompson, 2018)—has been acknowledged, detailed exploration into these areas remains scarce.

Several studies have examined the relationship between individual capabilities in human resources and aspects such as organizational culture, leadership, and motivation within larger business contexts (Schein, 2016; Tannenbaum, 1966). However, these investigations have predominantly focused on larger, publicly traded companies, with startups in the seed stage receiving limited scrutiny. Although some surveys have included small and seed-stage startups, their quantitative analyses usually encompass more mature companies, restricted to scale and growth stages (Men, 2021). Moreover, despite some research on entrepreneurship's link to leadership absence (Girdauskiene et al., 2015), there is a lack of studies differentiating startups by their growth stages and how specific deficiencies affect them. A notable reference (Gbadegeshin et al., 2022) aggregates research on risk factors leading to startup failure without distinguishing among the startup phases.

Consequently, there is an urgent need for a unified framework capable of systematically assessing and identifying the competencies related to leadership and organizational core strengths during the seed stage of high-tech startups.

Prior research has shown that uncertain events are common in seed-stage startups.

Startups attempting to innovate through more advanced technologies face higher uncertainty levels (Borracci et al., 2021; Byrne et al., 2020; Felsman et al., 2020), contributing to stress and rapid organizational changes (Tannenbaum, 1966). Therefore, in a startup with a small number of employees in the seed phase, employees as well as the Founder are likely to be exposed to a high level of uncertainty, and in order to achieve high performance in the face of such uncertainty, tolerance to these rapid changes and uncertain events is considered to be related to competencies.

Purpose

Recruiting initial team members is crucial, as they significantly influence the company's culture. During the seed phase, management often grapples with resource constraints, impacting training, personnel evaluation, and critical hiring and termination decisions. Additionally, there is a pressing need for an effective method to gauge potential recruits' alignment with the company's culture and characteristics, and their ability to take on leadership roles and form the organization's nucleus. The author also hypothesized that startup employees are exposed to high uncertainty during the seed phase as mentioned earlier, and that their tolerance for uncertainty has a lot to do with their competencies in order to achieve high performance in this environment.

Thus, this study poses a research question: Are the abilities of tolerance for uncertainty crucial for individuals to form the core of an organization, inspire fellow team members, and contribute substantial value?

Our primary goal is to develop a framework that streamlines the identification of these essential competencies. This framework will facilitate their incorporation into human resource strategies, including recruitment and training, specifically designed for high-tech startups in the seed stage.

The ultimate objective of our research is to create a methodology that enables seed-stage startups to make informed hiring and firing decisions more efficiently. This is particularly vital for organizations with limited resources for these processes but are keen on expanding their organizational scale through strategic recruitment.

2 Prior Research

Tolerance for Uncertainty

Many studies have used the concept of human uncertainty tolerance (Dantan et al., 2013; Lawrence & Lorsch, 1967; Lubart, 2016; Thompson, 1967).

The study in Lubart (2016) investigated uncertainty tolerance, personality traits, and their relationship to specialty selection among medical students considering various aspects of their careers (risk, ambiguity, complexity). Students with high uncertainty at the beginning of their studies show reduced tolerance over time, while those with low tolerance tend to increase it after enrollment. However, the cause of these changes remains unclear and further interventions may have affected the results.

On the other hand, studies that took a psychological approach include (Dantan et al., 2013) and (Thompson, 1967), which studied the influencing factors.

The research in Dantan et al. (2013) posits that the ability to think divergently, the polar opposite of convergent thinking observed among patients suffering from mental health disorders such as depression, is closely linked to uncertainty tolerance. The appropriateness of interventions such as improvisational theater as a therapeutic method for individuals unable to think of alternatives in response to a single event was investigated. Divergent thinking has been associated with success in creative fields such as entrepreneurship (Lawrence & Lorsch, 1967). Hence, in Thompson (1967), the authors medically examined the effects of changes in saliva volume on the ability to make decisions under uncertainty and risk and found that acute stress improved decision-making skills and increased tolerance levels.

Although methodological studies have used uncertainty as a probability distribution (Knight, 1921), creating a probability distribution model for each human statement or situation is challenging, and no such previous studies exist.

Studies have shown that the external environment of management organizations is full of complexity and uncertainty (Bhattacharya et al., 2017; Iyke, 2020).

Based on these previous studies, uncertainty tolerance in seed stage startups may be associated with the ability to think divergently while improving decision-making.

Because these are related to entrepreneurial success, uncertainty tolerance may also be important in the seed stage high-tech startups, which are the target of this study.

In addition, as discussed in Sect. 1.1, prior studies such as (Borracci et al., 2021; Byrne et al., 2020; Felsman et al., 2020) have shown that uncertain events are common

in seed-stage startups, and employees as well as founders are likely to be exposed to high uncertainty. Therefore, these characteristics are very important for seed-stage startups where the overall performance of the employees is likely to be directly related to the company's performance.

Method of Analyzing Human Resource Capabilities in a Company

Previous studies have used text mining to analyze the connection between motivation, leadership, and company results, using meeting minutes, daily reports, and other data (Blair et al., 2014). The method used in these studies is to conduct text mining first, followed by quantitative analysis, and finally, qualitative evaluation of the results. Some studies using text mining have been conducted on startups (Bzhalava et al., 2018; Ilya et al., 2018; Sung-Soo & Yang, 2017), but most are limited to the analysis of startup profiles and portfolios, whereas some use text mining to delve deeper into the characteristics of individual organizations, such as large firms.

It is desirable to generalize the conditions of these studies and create a standard framework using an actual analysis rather than through case studies. The population is assumed to be similar, even if the dates and times of meetings differ. Therefore, we will use the method in (Ishida & Kin, 2012) and other methods to apply the parameters obtained from the quantitative analysis to other data groups.

3 Methodology

Research Question

For small organizations, it is essential to be involved with the external environment while making managerial and recruitment decisions. The organization will be structured considering various uncertainties, including the company's survival.

In (Dantan et al., 2013; Lubart, 2016), and others, the ability to think divergently is considered as the ability to tolerate uncertainty and high risk. Moreover, in Lubart (2016), it is shown that this ability leads to entrepreneurial success.

In this study, the authors hypothesize that the ability to tolerate uncertainty is related to the research question of what the necessary human resource strategies of a high-tech startup organization in the seed stage are. Therefore, the research question for this study is that uncertainty tolerance is deeply linked to competence in startups. The hypothesis for quantitatively examining uncertainty tolerance in this research question is based on the previous literature (Thompson, 1967), which suggests that uncertainty tolerance can be examined by examining whether conversations are not too focused on a single topic or a single word, and whether they are moderately

divergent. Therefore, the authors hypothesized that examining the divergent nature of conversations during interviews could lead to an assessment of uncertainty tolerance and competency in startups.

Planning the Analysis Objectives

We noted in the previous section that uncertainty tolerance as a hypothesis of competency in seed stage startups may be manifested in whether or not they converge on a single topic or word in a conversation.

To test this hypothesis, we thought we could examine some of the differences in uncertainty tolerance by analyzing the differences between employees who continue to perform well in seed-stage startups and those who leave because they do not perform well.

Since uncertainty abounds on a daily basis, we can examine uncertainty tolerance by examining the text of how people speak when uncertain events occur in their daily conversations. Here, recordings of meetings at Adansons Inc, which the author manages and conducts action research for, will be used for analysis.

When selecting targets for analysis, special consideration will be given to the objective of comprehending how day-to-day operations tolerate uncertainty and the outcomes that ensue. Recordings of online meeting tools are selected and used for analysis at the most appropriate meetings during the action research. Adansons Inc. possesses several patents for technologies that serve as the foundation for Tohoku University's innovations. These technologies are exceptionally advanced, boasting a multitude of potential applications that necessitate corresponding theoretical extensions. Additionally, they encompass elements and potentials that remain undetermined. From a management perspective, this technology stack ranks among the most uncertain. Therefore, Adansons Inc. has a committee to understand these technologies, find the correct usage of each technology in different areas, search for technologies that are currently lacking, and discuss the possibility of their business applications, including formulating a direction, sharing results and knowledge, and implementing in-house demonstration for various use cases. Uncertainties in the technology and market must be considered to determine whether a technology is inapplicable in a particular area or can be applied through technological expansion. The content of the statements includes decisions about considering uncertainties in formulating the company's next course of action. Furthermore, this committee had members who participated only in January 2021–September 2021 and those who subsequently left the company due to disagreements with the organizational culture. We analyze the differences in characteristics between members central to the organization, other members, and members who have left the organization from the perspective of what kind of differences in speech characteristics led to a sense of disagreement with the organizational culture.

There were five participants in this committee; two managers and three engineers, one of the engineers being an employee who had left the company due to disagreements in thinking and culture. The characteristics of the five participants, referred to as Subjects 1–5, are as follows:

Subject 1: Employee, engineer. From 2020 to the present, he has been highly central in the organization and is in a position similar to that of a manager. Although he has theoretical knowledge, he actively studies what he does not know and tries to make the most of it. The author's experience before the analysis is that Subject 1 can formulate his own hypotheses and opinions, not limited to this meeting group, on unknown events that are often unknown even to the author and his management team.

Subject 2: Former employee, now retired. Engineer. He has felt friction since the committee was formed due to discrepancies in the direction of conversations and ideas. Although unfamiliar with methods or patents, his mathematical and programming skills are the best in the group. In technical validation, he would often independently work with the content, performing tasks outside what he had been instructed, but sometimes went off the rails due to differences in thinking.

Subject 3: An engineer and tech lead who is still active. Like Subject 2, he has good mathematical and programming skills and is most familiar with methods and theories related to patents. Moreover, as a Tech Lead, he is involved in developing new theories and technologies within the organization. The author's experience prior to the analysis is that Subject 3 can see the big picture, which allows him to take conversations smoothly. In addition, because of his ability to see the whole picture, he stays calm and within the general framework, even when doing something outside the instructions during the technical verification process.

Subject 4: The board member, author, and CTO who moderated this meeting. As the CTO, he has a deep understanding of patented technologies. However, his role is to control the overall direction of the meeting so that the discussion stays on-topic; hence, his methodological comments are less frequent than those of Subject 5.

Subject 5: Executive, product manager. In this committee, he is central in explaining the patented technology, discussing aspects other members may overlook, and providing direction to the employees, including technical comments on programming. He has the deepest understanding of patented technologies in the company and is well-versed in the theories necessary to extend the technology.

The meetings occurred on a weekly basis, with occasional absences by one of the five members. All meetings were recorded through both video and audio using an online conferencing tool.

The focus of the meetings, which forms the basis for our analysis, encompassed discussions where two members of the management team, possessing expertise in patented technologies, imparted knowledge to other team members on how to utilize these technologies effectively. Additionally, each member explored strategies for applying patented and related technologies to specific use cases with the aim of enhancing their effectiveness and practicality. The meetings were structured around the objective of instructing employees in how to approach the utilization of patented technologies, as well as how to analyze and evaluate data within this context. While

the two management team members possessed the knowledge of how to navigate such scenarios, there were numerous aspects open to discussion, primarily due to the inherent uncertainty associated with these endeavors. Moreover, discussions focused on interpreting the results obtained through the use of patented technology, acknowledging that such outcomes can often be subject to multiple interpretations. Consequently, this meeting group was selected as the subject of analysis, as it aligns well with the research question's objective, fostering discussions amidst a backdrop of uncertainty. Additionally, the weekly meeting cadence imposed a constant need to formulate actions for the subsequent week, even in the face of uncertainty.

It is important to note that there were only five participants and only one retired subject. Therefore, it is possible that there may be a tendency or bias specific to that one person. However, this study will be reported as a case study, examining the relationship between the characteristics and competencies possessed by the person who retired as a matter of fact.

In addition, my personal feelings and biases should be eliminated as much as possible. Therefore, quantitative analysis methods such as PCA were used to determine whether retired subjects had different tendencies on a numerical basis, so that the results of the study would not differ due to biases beyond necessity.

Text Mining

The premise of this study is that it is a case study of a single company, and by generalizing it, we aim to make it applicable to other companies. Therefore, the analysis will be qualitative and not quantitative. However, generalizations will be made regarding possible conditions and environments so that the analysis is not limited to the case study category.

As an analysis method, we visualize the meeting content, including spoken words, and convert the recorded data from the meeting into a form that can be analyzed by text mining to generalize the object of analysis. The use of text mining in such analyses is common, and numerous similar prior studies exist (Bzhalava et al., 2018; Ilya et al., 2018; Sung-Soo & Yang, 2017).

When performing text mining, the data is separated into words and phrases, the frequency of occurrence of each word is calculated, and the list of extracted words is created as a dataset. KHCoder (Corley et al., 2007) is used to create the list; the number of occurrences is recorded after morphological analysis and segmentation.

During word extraction, it is common to narrow down to a certain part of speech or exclude exclamations, especially when looking for common characteristics. However, words related to stilted speech, such as "ah" and "e," are strongly remembered, and although not meaningful by themselves, they make it difficult to predict the word immediately following them. Moreover, fluent and non-fluent speakers differ in the ease with which they convey their conversations, including parts of sentences that make others strongly aware of them (Hlavac, 2011).

In contrast, slowness and sentence breaks are important in clarifying the speaker's feelings (Clark et al., 2002) and can lead to expressing intentions such as the desire to transfer the right to speak to another (Ewing et al. 2017).

In this study, we assumed that it is unclear what characteristics of uncertainty tolerance are expressed in words and what trivial behaviors other than the purpose of conversation may be involved in such characteristics. Hence, we treated all extracted words as equally important and analyzed them without restrictions in terms of parts of speech.

Quantitative Analysis

Quantitative analysis was performed on the list of extracted words. In this study, the analysis focused on the behavior of retired Subject 2 compared to that of the other subjects.

As a method for quantitative analysis using text mining, a co-occurrence network was first used to visualize the frequency of occurrence of words and the relationship between subjects, and then a qualitative evaluation is conducted. A co-occurrence network is a diagrammatic representation of commonalities between words in a sentence and is a well-known text mining technique. To create the co-occurrence network, the Jaccard coefficient calculates similarity in texts based on the frequency with which two words appear simultaneously; the higher the coefficient, the higher the similarity. The Jaccard coefficient is a measure of the similarity of sets, and in text mining, it is a measure of the similarity, or distance, between sentences. Weights are then calculated considering which words appear more frequently with which speakers; the higher the weight, the thicker the line between speakers and words in the plotted co-occurrence network. Thus, by visualizing the co-occurrence network, it is possible to observe objectively which speaker utters which words more often. Moreover, co-occurrence networks assume that the speaking habits of each person are also meaningful in expressing the characteristics and personality of each person, and analysis is performed including all parts of speech that can be included in KHCoder's morphological analysis, such as adverbs, emotional verbs, and suffixes. The top 100 words were used to draw co-occurrence relations.

After visualizing the co-occurrence network, a quantitative analysis method examines the words uttered and speaker characteristics.

Quantitative text mining analysis includes correspondence analysis and principal component analysis (PCA) (Greenacre, 2017). Correspondence analysis is widely used as it can easily visualize the relationship between categories (Ihaka & Gentleman, 1996). However, it does not consider the number of word samples, making it difficult to measure which words contribute significantly to separating specific speakers or identifying characteristics. In contrast, PCA considers the number of samples and analyzes the relationship between each principal component and the speaker using contribution analysis to each principal component. PCA

searches for specific cohesion in cluster analysis when the number of samples is sufficiently large and quantitative analysis is possible. However, in this study, a sample size based on the number of utterances per meeting can be considered a sufficient sample size. Therefore, PCA, which is more advanced than correspondence analysis, was adapted to qualitatively analyze the characteristics of the entire meeting based on the overall results.

PCA was performed using Scikit-learn (Rossum & Drake, 2009), a library in Python (Pedregosa et al., 2011), which runs PCA on a tabular dataset that records the number of words uttered by each subject on the vertical and horizontal axes. The values were standardized by subtracting the mean of all utterances from the number of times which speaker uttered which word and dividing it by the standard deviation so that the mean was 0 and the standard deviation was 1. The reason for using standardization rather than normalization is that the upper limit of utterance for each word at each meeting session is different. Similar treatment of various parts of speech may result in large differences for each part and consequently bias the results for one part of speech. PCA provides a vector of scores for each principal component for each speaker, displayed simultaneously with the vector of each word on a vector space based on each principal component, to visualize which speakers have what word component strongly, even when clustering by word is difficult. The relationship between the principal components and the corresponding score vectors was then examined to investigate what characteristics differentiated Subject 2 from the other subjects.

Also, when performing PCA, it is expected that there will be multiple principal components. Since this is a meeting where all participants are discussing a common agenda, there is a possibility that words related to the agenda that make up the majority of the conversation may appear specifically in the first principal component, etc. If a global principal component appears that has a contribution rate of over 70%, it shall be excluded from the analysis and the remaining components shall be analyzed. The principal components used in the analysis shall be selected according to the number of participants in the meeting being analyzed. For example, if there are five participants, up to the fifth principal component shall be analyzed, and if there are four participants, up to the fourth principal component shall be analyzed. However, the number of principal components for which information related to the subject being tracked will vary from one analysis data to another and cannot be determined in advance. Therefore, we will utilize principal component vectors to calculate correlations to determine which subject's component is dominant in which principal component each time.

Dataset

In this study, we used "Whisper," a large language model created by OpenAI, to perform highly accurate automatic transcription (Radford et al., 2022). In recent

years, deep learning-based large language models have become a mainstream transcription method. Whisper was selected for this analysis because the pre-trained model was free and easy to use, and the documentation on how to use it is publicly available and widely recognized by hightech venture companies. However, speaker annotation and transcription error corrections were made manually.

For the analysis, three meetings with distinctive features were chosen: The first meeting took place on July 22, 2021. During this session, the executive provided a unilateral explanation to the employee regarding the meeting's objectives and the rationale behind task assignments. This meeting also included a brief question and answer session. The second meeting, held on August 19, 2021 involved progress reports from each employee (excluding the executive) and instructions from the executive regarding necessary actions and findings.

The third dataset was derived from a meeting that took place on August 26, 2021. Similar to the second dataset, this meeting involved employees presenting progress reports, followed by mutual feedback, including input from the executives. Following this, employees expressed their opinions regarding necessary actions. The author's selection comprised three types of meetings: those in which instructions were the primary focus, meetings in which employees both reported their opinions and provided instructions, and meetings where the emphasis was placed on employees' opinions.

One caveat to this dataset is that there is only one retiree, and it is possible that characteristics specific to that retiree will be perceived as distant from competency. However, if the person who actually retired has characteristics that are different from others, it is necessary to identify those specific characteristics in this case study. On the other hand, considering the possibility that the author has a bias that the retired person has characteristics different from others that are idiosyncratic, we have tried to eliminate that bias and enable a neutral evaluation by combining quantitative methods as well as qualitative evaluation.

4 Analysis

Word Extraction Results by Morphological Analysis

The results of the occurrence counts are shown after morphological analysis and sharpened processing were performed using KHCoder. The analysis was performed for each of the three datasets.

The extraction results showed that the first dataset had 217 sentences and 129 utterances, containing 10,090 words, with 1228 different words. The second dataset had 259 sentences uttered 121 times, of which 11,258 words were extracted, with 1259 different words. The third dataset had 262 sentences and 121 utterances, containing 7457 words, with 972 different words.

Co-occurrence Network

Figure 1 shows the results of displaying the co-occurrence network for the first meeting. The co-occurrence network provides a visual display of the relationship between words and speakers. However, to convert this qualitative study to a quantitative study and eliminate observer bias, PCA, described below, should be used.

A tendency to use more technical terms is observed in Subjects 1 and 2. In addition, all but Subject 2 use the word "do" often.

Figure 2 displays the co-occurring network for the third dataset. Notably, Subject 2 frequently referenced specific names of analysis methods, whereas Subjects 1, 4, and 5 employed words suggesting vagueness, such as "something," more frequently. Observations from the co-occurrence networks across all three meetings reveal that Subjects 2, 3, and 5 frequently employ technical terminology, including method names unrelated to patented technology. In contrast, Subjects 1 and 4 utilize words beyond method names and exhibit connections to all other subjects through these words.

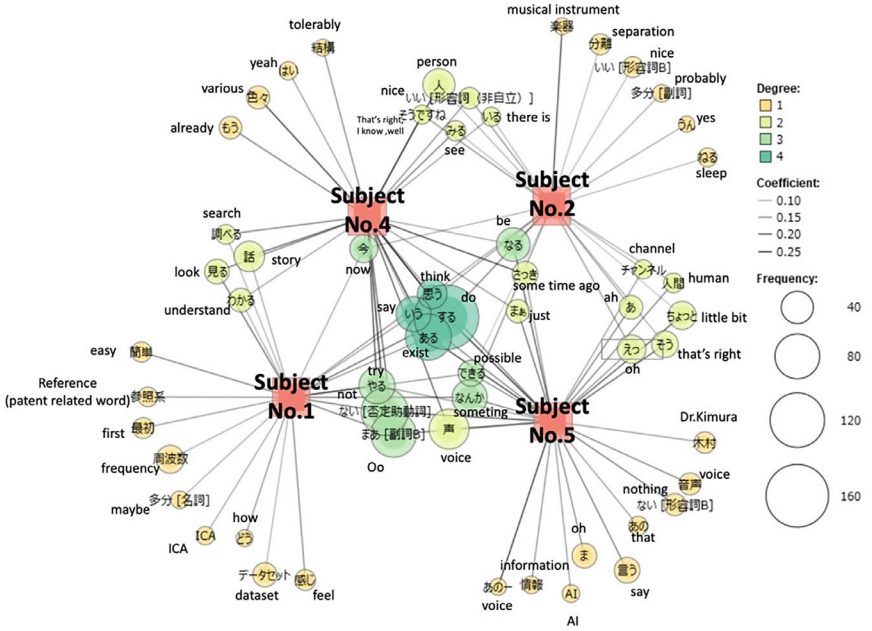


Fig. 1 Co-occurrence network for the July 22, 2021meeting

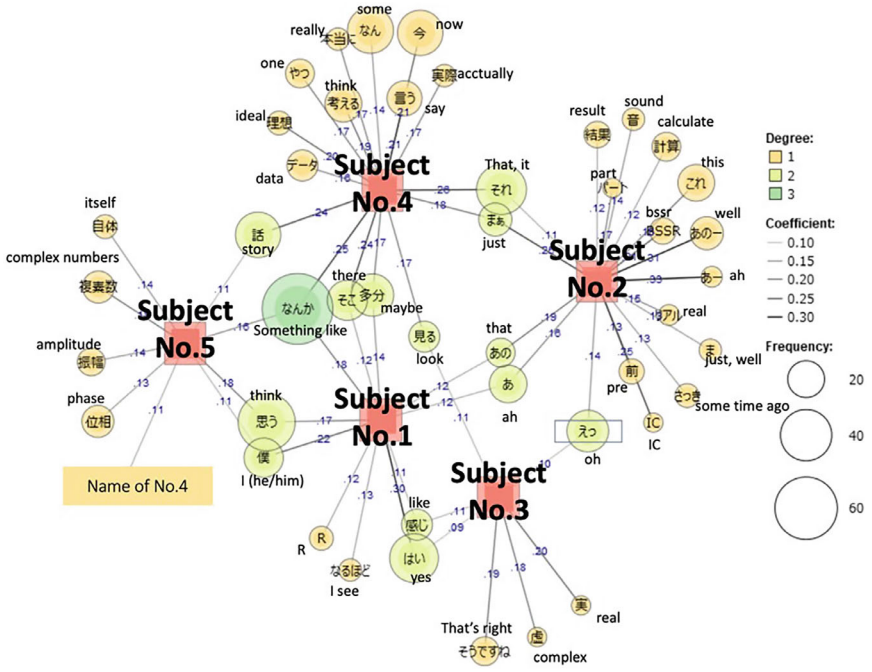


Fig. 2 Co-occurrence network for the August 26, 2021 meeting

Principal Component Analysis

We begin by providing the results of the PCA conducted for the initial meeting. A table was created, featuring subject names along the vertical axis and words along the horizontal axis, as illustrated in Fig. 3. Following the construction of this matrix, PCA analysis was performed.

The principal components were decomposed into four groups, corresponding to the number of participants. First, Fig. 4 displays the principal component scores, where each principal component's name is depicted on the horizontal axis, and subject numbers are presented on the vertical axis. The vertical axis quantifies the contribution of each principal component when expressed in terms of the four principal components, facilitating the identification of patterns and associations within each. Additionally, PC1 and PC2 are indicated sequentially, starting with the first principal component.

The calculated contribution ratio of each principal component and cumulative contribution ratio are shown in Figs. 5 and 6, respectively.

	talk	something	easy	What?	Reference (patent related word)	area	me	various	...
Subject4	1.477098	-1.214286	0.055556	-0.648901	-0.405751	0.150756	-0.160817	1.459993	...
Subject2	-0.738549	-0.357143	-0.611111	0.034153	-0.676252	-0.753778	-0.804084	-0.811107	...
Subject1	-0.369274	1.071429	1.388889	-0.785512	1.487755	1.356801	1.447352	-0.324443	...
Subject5	-0.369274	0.5	-0.833333	1.400261	-0.405751	-0.753778	-0.482451	-0.324443	...

Fig. 3 Part of the data standardized for the number of utterances at the meeting on July 22, 2021

	PC1	PC2	PC3	PC4
Subject4	-2.846651	5.528277	-2.01736	-1.33E-15
Subject2	-4.677224	-2.040037	4.103563	4.09E-15
Subject1	0.163256	-4.428146	-4.002342	-7.45E-15
Subject5	7.360619	0.939906	1.916139	4.69E-15

Fig. 4 Principal component scores when principal component analysis (PCA) is applied to the July 22, 2021, meeting

PC1	4.68E-01
PC2	3.07E-01
PC3	2.26E-01
PC4	2.71E-33

Fig. 5 Contribution of PCA when applied to the July 22, 2021, meeting

Almost 100% of the information is explained by up to PC3. A portion of each principal component vector calculated using Scikit-learn's pca.components is shown in Fig. A1 of the Appendix Section. The vertical and horizontal axes represent each principal component name and word, respectively.

To analyze the relationship between each principal component and subject, we calculated the correlation coefficient between the matrix showing the number of utterances of each word for each subject (Fig. 3) and the principal component vector matrix showing the relationship between each principal component and word calculated using PCA (Fig. 7). Correlation coefficient is an index that measures the strength of a linear relationship between two sets of data and takes a value between -1 and 1

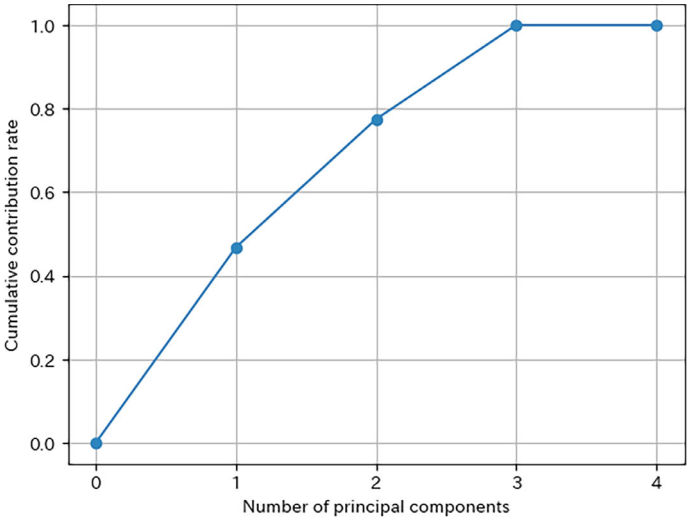


Fig. 6 Cumulative contribution of PCA applied to the July 22, 2021, meeting

as a real number. 0 is uncorrelated. And the larger the value, the stronger the positive correlation, i.e., the higher the relationship. The results in Fig. A2 of the Appendix Section show the color map created based on the maximum and minimum values of the correlation coefficients. The larger the value, the higher the correlation coefficient, and the greater the positive correlation, the darker the color. A correlation coefficient between 0.7 and 1 is considered a strong positive correlation (Ratner, 2009).

Figure A3 of the Appendix Section shows the components of each principal component vector, including PC1 having a particularly high positive correlation only for Subject 2, and words that are positive and exceeding the thresholded value of 0.1 for each principal component (Figs. 7, 8 and 9).

Next, we show the relationship between words and speakers when using correspondence analysis. However, due to the small number of samples and the large variance, it is difficult to perform clustering. Therefore, the principal component score vector for each speaker obtained using PCA is displayed simultaneously with the vector of each word on a vector space based on each principal component to visualize the relationship between each speaker and word, as well as distribution bias. Figures 10, 11 and 12 show the relationship between each principal component and word in a scatterplot.

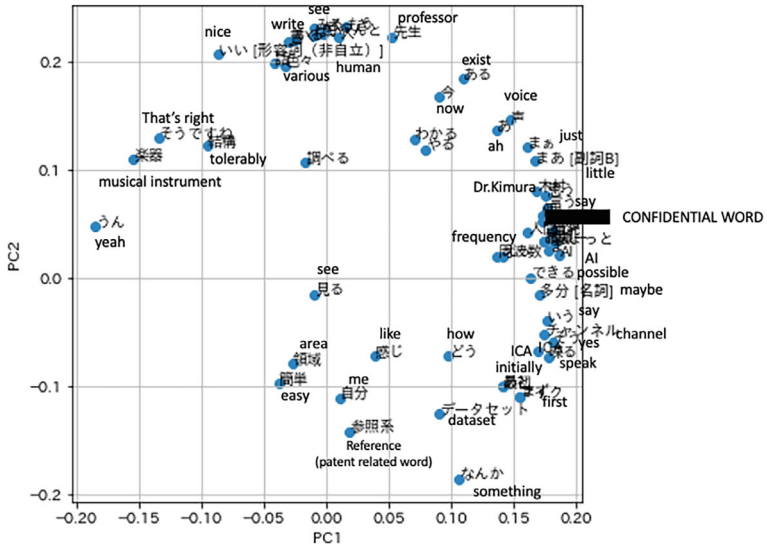


Fig. 7 Mapping of each word on the space defined by PC1 and PC2 when PCA was conducted for the July 22, 2021, meeting

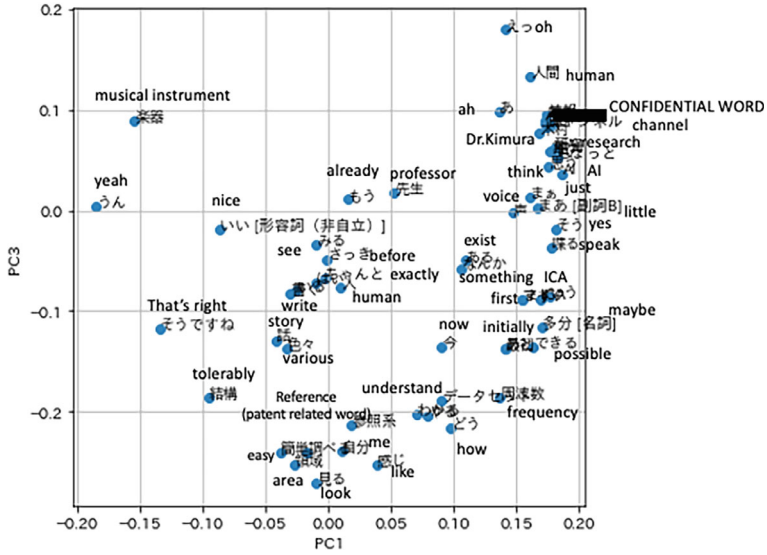


Fig. 8 Mapping of each word on the space defined by PC1 and PC3 when PCA was conducted for the July 22, 2021, meeting

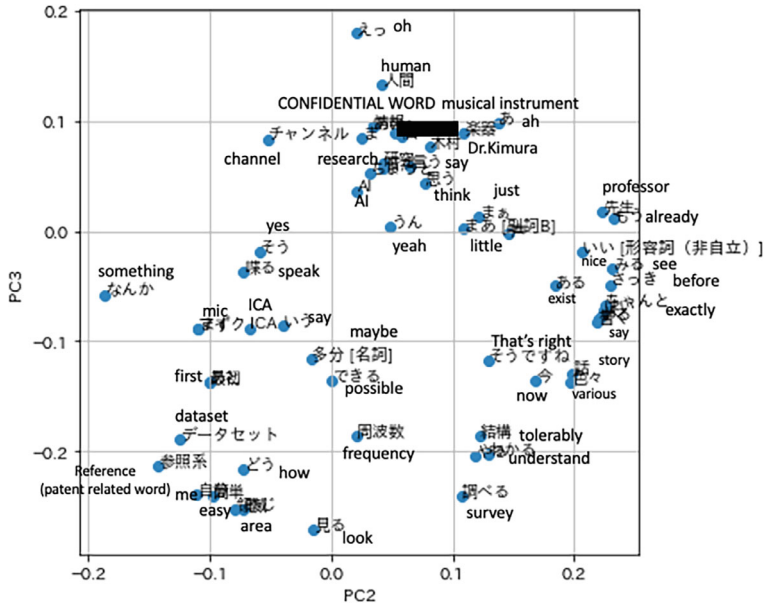


Fig. 9 Mapping each word on the space defined by PC2 and PC3 when PCA was conducted for the July 22, 2021, meeting

Next, the principal component score vectors for each speaker obtained using PCA are visualized by displaying them simultaneously with the vector of each word on a vector space based on each principal component, as shown in Figs. 13, 14 and 15. Principal component scores are the values the data take on each principal component axis.

If the number of participants is large and difficult to separate, the analysis can be extended to multiple dimensions, as shown in Fig. 16.

These procedures were followed for the two meetings in August. Below are excerpts of the characteristic plots from each meeting. Figures 14 and 15 show the mapping of each word on the space defined by PC2 and PC3 for the August 19, 2021 meeting and the plot of principal component vectors. Figures 16 and 17 show the mapping of each word onto the space defined by PC2 and PC4 for the August 26, 2021 meeting and the plot of the principal component vectors.

The analysis results for all meetings showed that Subject 2 was characteristically isolated, with a strong correlation to technical terms.

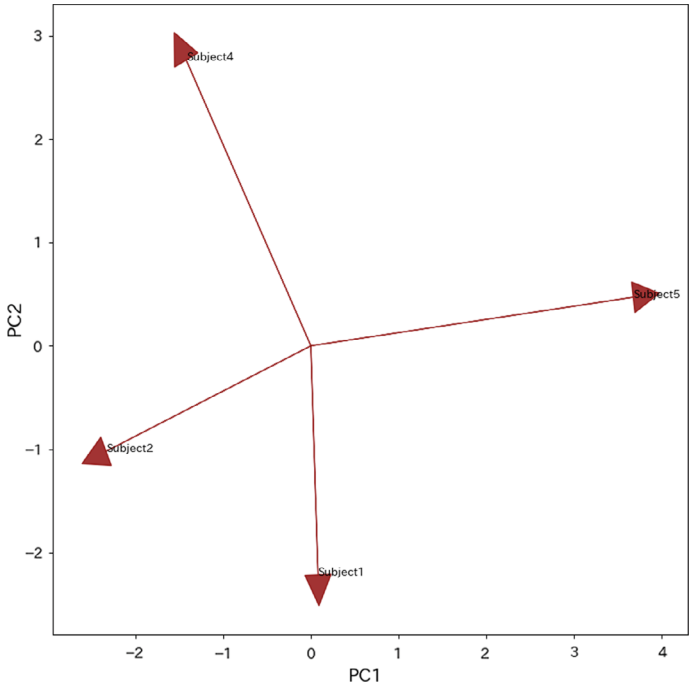


Fig. 10 Principal component vectors for each speaker on the space defined by PC1 and PC2 when PCA was conducted for the July 22, 2021 meeting

5 Discussion

Features from Quantitative Analysis

The PCA results reveal that Subject 2 exhibits distinct components compared to the other subjects. In this section, we delve into the interpretation of these components derived from each analysis.

Initially, it is observed that the components strongly correlated with Subject 2 are either uncorrelated or exhibit a pronounced negative correlation with the other subjects. Consequently, the relevant principal components offer the capability to distinguish Subject 2 from the rest. As such, we will conduct a detailed examination of the words closely associated with Subject 2 and their characteristics. This assessment will consider the frequency of words that might not have been adequately addressed in the correspondence analysis.

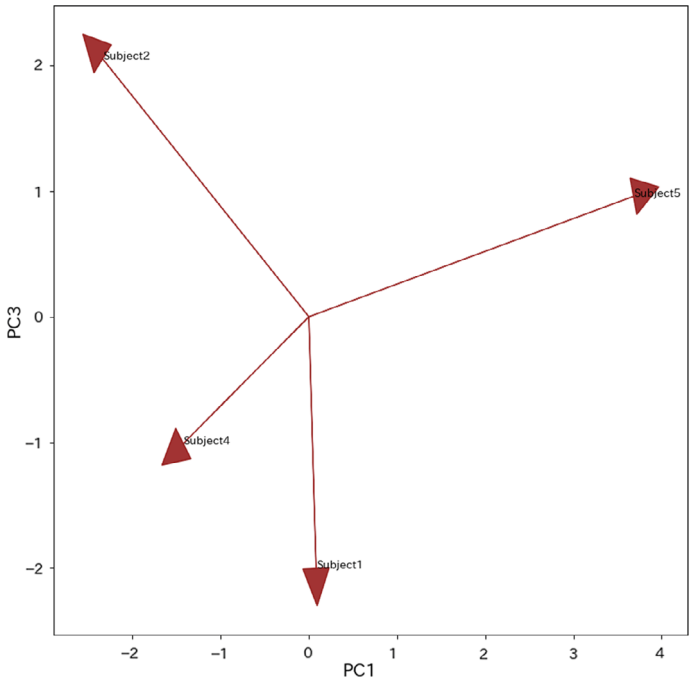


Fig. 11 Principal component vectors for each speaker on the space defined by PC1 and PC3 when PCA was conducted for the July 22, 2021 meeting

In the case of the meeting held on July 22, 2021, PC1 displays a high correlation coefficient of 0.85 with Subject 2, with a notable emphasis on terms related to patents and technical jargon. Technical terms are distributed across PC1 to PC3, with PC1 encompassing words used by Subject 4 to elucidate concepts to other participants, PC3 containing technical terms associated with patents, and PC2 encompassing other technical terminology. Furthermore, during the August 26, 2021 meeting, Subject 2 exhibits a strong correlation with PC4, with a coefficient of 0.97. This is particularly conspicuous in the case of technical terms unrelated to patents.

These findings suggest that Subject 2 employs a more extensive array of technical terminology compared to the other subjects, and these characteristics are closely associated with the use of technical terms. Furthermore, it is notable that technical terms, even those not directly linked to patented technology or the central focus of the conversation, may occur more frequently in Subject 2's contributions.

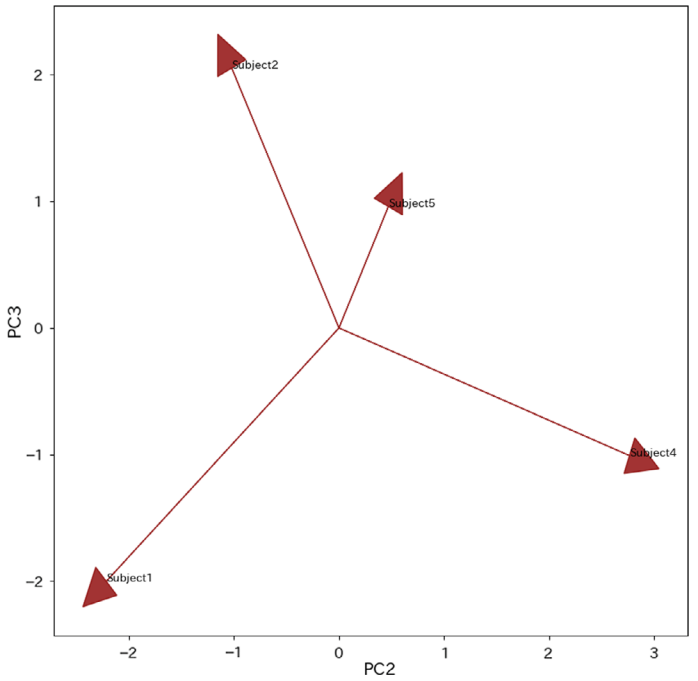


Fig. 12 Principal component vectors for each speaker on the space defined by PC2 and PC3 when PCA was conducted for the July 22, 2021 meeting

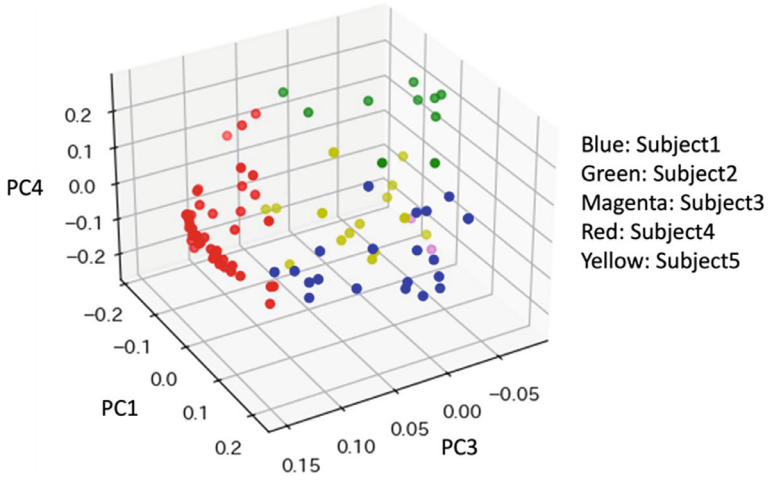


Fig. 13 Scatter plots for PC1, PC3, and PC4 based on principal component analysis

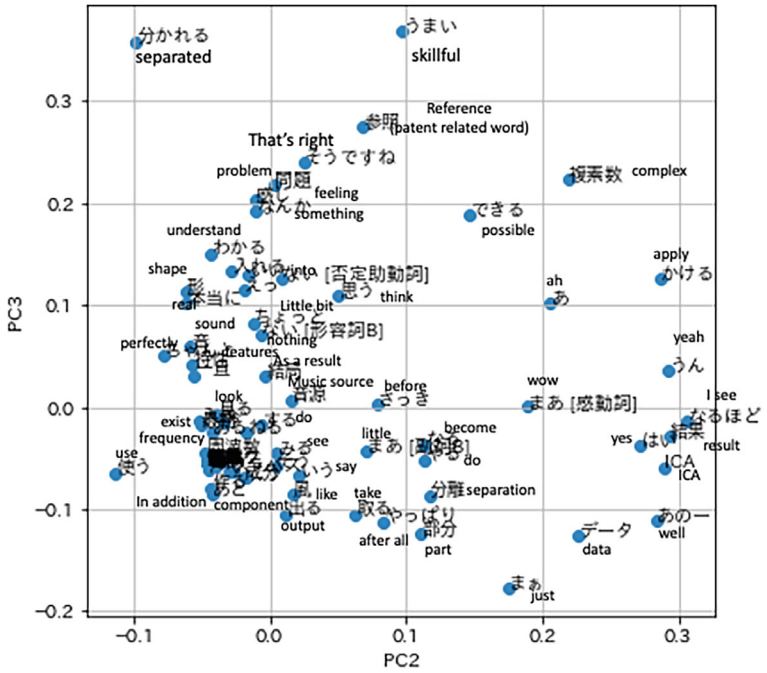


Fig. 14 Mapping of each word on the space defined by PC2 and PC3 when PCA was conducted for the August 19, 2021 meeting

Relationship to Uncertainty Tolerance

Prior research has shown that uncertainty tolerance is largely related to divergent thinking and does not converge in response to uncertainty and risk types (Lubart, 2016; Matta et al., 2022). However, there is a difference between Subject 2 and other subjects in terms of whether there were more words with technical terms for the patented technology whose usage was non-trivial. In addition, discussions using technical words not directly encompassed by the patented technology were particularly noticeable. The qualitative evaluation reveals that Subject 2 engaged in discussions on methods to verify the patented technology, unlike the purpose of the meeting, which was to understand how to use the patented technology and explore its application areas.

The nature of the discussion may be considered similar to convergent thinking in prior studies. Thus, more discussions using technical terms than business-like or schematic discussions are possible.

In addition, such discussions of technical terms occur not in progress reports and the like but rather when formulating what to do next, the so-called Next Action. The ability to think about the next actions during uncertainty and the rate at which methodological conversations are uttered may have contributed to this. Therefore,

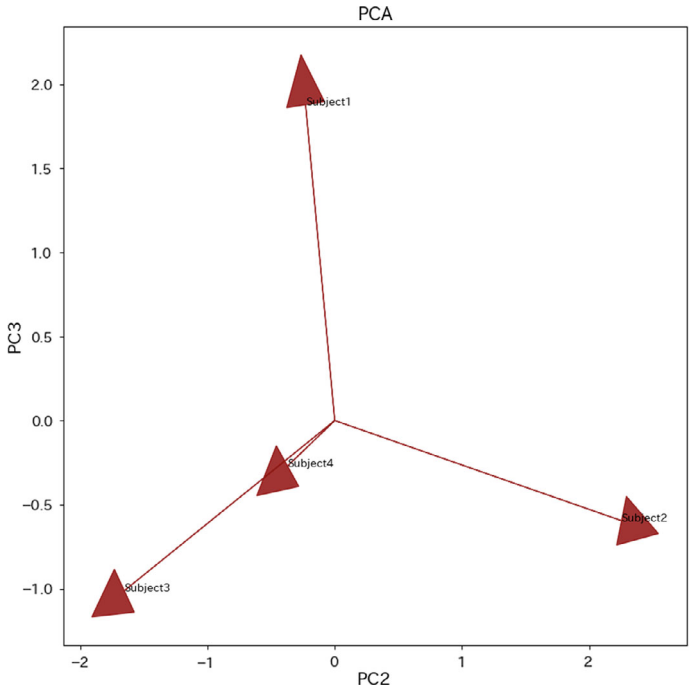


Fig. 15 Principal component vectors for each speaker on the space defined PC2 and PC3 when PCA was conducted for the August 19, 2021 meeting

the ability to tolerate uncertain content means the ability to make assumptions as well as facts in the face of uncertainty and be aware of the fundamental premise without being bound by the methodology.

Challenges in Utilizing the Method as a Framework

This section explores the applicability of the findings from this study to other meetings and organizations, along with associated challenges.

Using the components derived from PCA, it is possible to project these components by taking the dot product of eigenvectors from the variance–covariance matrix, as exemplified by the August 26 meeting, and comparing them with those obtained through PCA in similar analyses of other meetings. Variance–covariance matrix is a matrix representation of the covariance between the elements of a vector. It is an extension of the concept of variance in a scalar-valued random variable to multiple dimensions. This allows for the superimposition of principal component vectors. However, it is important to ensure consistency in the words used across meetings. Even when statements across multiple meetings share similarities in word usage,

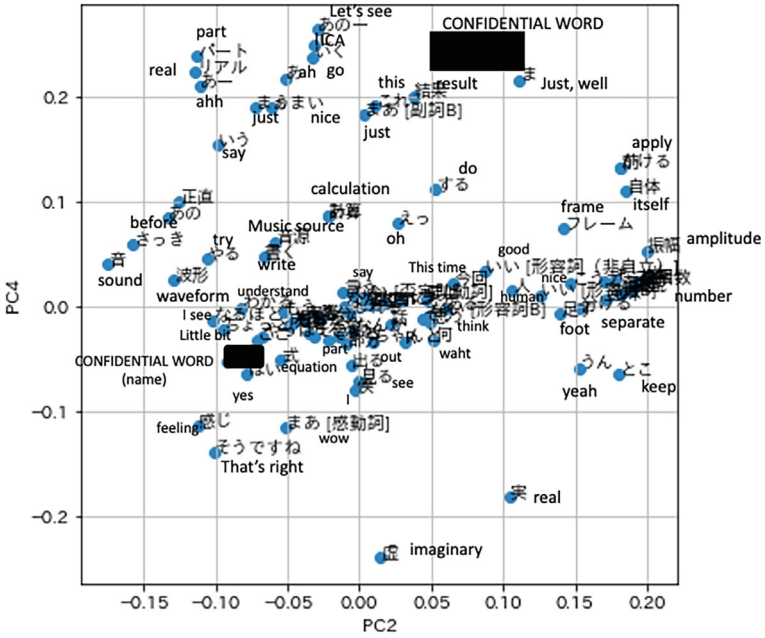


Fig. 16 Mapping of each word within the space defined by PC2 and PC4 when PCA was conducted for the August 26, 2021 meeting

there may still be differences, complicating the universal application of eigenvectors from the variance–covariance matrix in all cases.

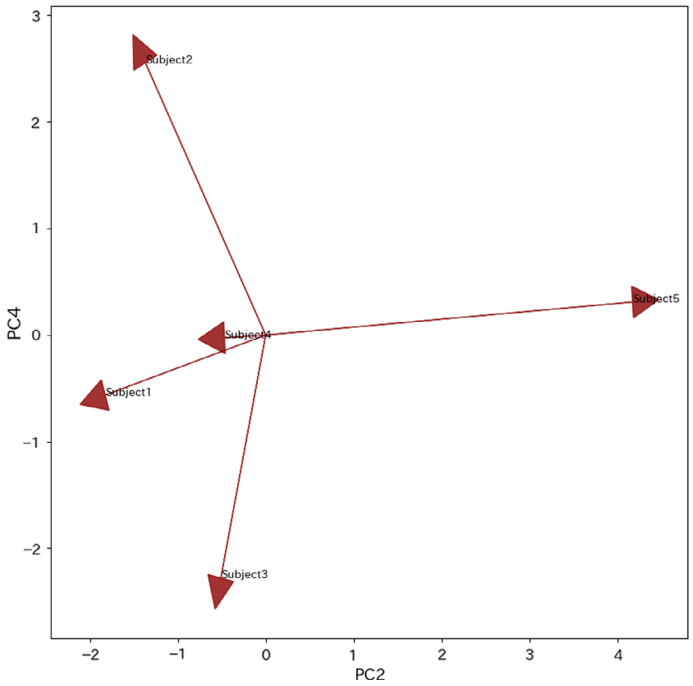


Fig. 17 Principal component vectors for each speaker within the space defined by PC2 and PC4 during PCA conducted for the August 26, 2021 meeting

The process of transcribing the three meetings, text mining to create a matrix of speakers and words, and using PCA to separate them is expected to be used as a framework. The separation of Subject 2 is reproducible in terms of visualizing using correlation coefficients, separating using a principal component vector and two principal components, and identifying components strongly related to the subject. The framework can detect differences between subjects by identifying components and analyzing words in the identified components. In addition, the framework could analyze speech tendencies over time, which could be transferred to an analysis of what kind of people may change due to their relationships with other people.

However, given the relatively small number of subjects compared to the data quadrants, it is challenging to rely solely on the visualization of principal component vectors for accurate separation. To establish a broadly applicable framework, expanding the dataset with more participants and meetings and confirming reproducibility under these conditions is essential.

Framework Applicability of Different Stages Startups and Other Industries

In Sect. 5.3, we discussed the versatility of the matrix calculation method to discover who tends to differ from others in their tolerance for uncertainty technically as a framework. In this section, we discuss how it can actually be applied to other stages of startups.

First, because of the importance of uncertainty tolerance in seed-stage startups, the analysis in this study was limited to seed-stage startups. In other stages of startups, especially in startups that have grown more than the seed stage, the positions that require uncertainty tolerance may be different. While this ability is not required of all employees as it is in the seed stage, it is very important for managers and others in positions that require them to take into account changing market needs and trends while hiring. In addition, in startups, it is as important as in the seed phase that it is very important for the company's growth strategy to be able to easily identify managers and other qualities at the time of hiring. Therefore, this framework could be applied to startups in other stages.

Moreover, this is not limited to startups. For example, departments that create new businesses, such as new business divisions of large companies, will face a great deal of uncertainty, despite the differences from startups. In such a situation, the survey could be applied as a framework to examine whether the discussion is not overly focused on a specific topic and whether it allows for uncertainty.

6 Conclusion

This study set out to answer the question: "What abilities are essential for individuals to become integral members of an organization's core group, inspire fellow team members, and demonstrate substantial value?" Our analysis focused on identifying the distinctive traits of individuals who have remained with the organization versus those who have departed. The findings revealed several key factors that may contribute to these differences, including the ability to analyze facts, think proactively amidst uncertainty, and the frequency of contributions to methodological discussions, as highlighted through Principal Component Analysis (PCA).

Previous research has indicated that a lack of essential employee skills poses a significant risk to startup viability (Barr et al., 2009; Cummings et al., 2018; Markham et al., 2010; Thompson, 2018), especially during the seed stage, which is characterized by notable fluctuations in organizational size and associated risks (Failed, 2021; Tomczak & Brem, 2013; Ycombinator, 2022; Zhang & Wang, 2017). Additionally, divergent thinking has been identified as a critical skill for entrepreneurs, closely tied to their capacity for ambiguity tolerance and creativity under conditions of high uncertainty and risk (Lubart, 2016). Existing frameworks have provided insights into strategic focus areas, such as strategy formulation, implementation, and resource

management during a startup's growth (Hu et al., 2021), emphasizing the importance of strategic leadership (Subramaniam & Shankar, 2020), strategic entrepreneurship (Morici & Zander, 2020), and adeptness in dynamic environments (Lubart, 2016; Schilke et al., 2018). However, there is a notable gap in qualitative case studies that demonstrate how these capabilities are crucial for preventing startup failures, particularly through the nonlinear and rapid application of technology. Moreover, the literature shows a lack of research on how these capabilities manifest in conversational attributes within startup organizations and the ways they can be quantified and operationalized within a framework (Lubart, 2016).

The theoretical implications of this study are significant. By focusing on a specific startup case study managed by the author, this research not only highlights the essential capabilities needed by startups (Hu et al., 2021; Lubart, 2016; Morici & Zander, 2020; Schilke et al., 2018; Subramaniam & Shankar, 2020), especially high-tech ones at the seed stage but also presents an operational framework for assessing startup uncertainties. This study emphasizes the critical role of uncertainty tolerance in startups and showcases how it translates into unique conversational traits. Our quantitative analysis, enabled by text mining, reveals that the ability to tolerate uncertainty is not merely about focusing on technical details through convergent thinking. Instead, it primarily involves divergent thinking, even among engineers, demonstrating that the ability to transcend technical jargon, hypothesize, stay mindful of basic principles without being limited by methodologies, and turn these insights into concrete actions is integral to engineers' divergent thinking abilities.

Practically, this study underscores the importance of accurately identifying such abilities in startup management and comprehending them through a framework that accommodates limited resources (Duane Ireland & Webb, 2007; Hitt et al., 2011; Hu et al., 2021). Text mining of employee conversations emerges as a valuable tool in this context. Furthermore, by visualizing the principal component vectors for two specific components derived, we can differentiate employees with varying characteristics, analyze their conversational traits, and employ this analysis as a framework. However, given that this analysis was conducted on a single company, its broader applicability to other companies with different environments and participant numbers remains a subject for discussion.

Looking ahead, there exists an opportunity for continuous analysis of these abilities within Adansons, Inc., using the developed framework. This could include monitoring how individual traits evolve over time, how they are influenced by others, and identifying convergent thinking tendencies during the hiring process to mitigate potential risks. The application of these frameworks could also aid in monitoring employee engagement trends and other relevant data, facilitating process consultation to tackle issues like unexplained resignations and improving recruitment and development practices. Additionally, exploring a simplified approach for analyzing characteristics through eigenvectors derived from the variance–covariance matrix via PCA, coupled with the dot product of this matrix and meeting data from various time points, could streamline the analysis process, enhancing its efficiency.

In addition, the sample size used in this analysis is very small and is a case study in a specific situation. While the intent of the study is to establish a methodology based on data from such a limited and specific situation, it is possible that specific biases may be included in the creation of the framework. The framework proposed here is only a possibility to analyze uncertainty tolerance in the general method (PCA), and is not dependent on the sample size of the present study. On the other hand, the very small sample size of about five participants may allow us to easily isolate the characteristic of uncertainty tolerance. In other words, in a meeting with more participants, uncertainty tolerance may appear more complex in the conversation, and caution should be exercised when applying this method in a meeting with more participants. However, PCA itself is a method that can handle an increase in the number of participants and the number of dimensions of the principal component vectors, and the flow of analysis proposed in this study itself can be considered to follow the same one. For more advanced analysis, it may be possible to increase the accuracy of PCA results by simultaneously utilizing methods to analyze components other than PCA. For example, combining with independent component analysis (ICA) may make it easier to find employees with peculiar characteristics, not limited to uncertainty tolerance, and we will consider combining such other methods with this framework in the future.

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Appendix

See (Figs. A1, A2 and A3).

	talk	something	easy	What	Reference (patent related word)	area	me	various	...
PC1	-0.041917	0.106697	-0.038094	0.140951	0.018701	-0.026493	0.010736	-0.154798	...
PC2	0.198506	-0.185796	-0.09743	0.020601	-0.14186	-0.0787	-0.110679	0.109422	...
PC3	-0.12907	-0.057786	-0.240778	0.17922	-0.214006	-0.25301	-0.238734	0.089002	...
PC4	-0.554806	-0.082988	0.042929	-0.137654	-0.171719	0.257574	0.116944	0.095412	...

Fig. A1 Each principal component vector when PCA is applied to the July 22, 2021, meeting

	Subject4	Subject2	Subject1	Subject5	PC1	PC2	PC3	PC4
Subject4	1.000000	-0.379487	0.406635	-0.003752	-0.760231	-0.435569	0.032495	0.947770
Subject2	-0.379487	1.000000	0.234999	-0.002168	0.850822	0.118890	-0.820452	-0.142587
Subject1	0.406635	0.234999	1.000000	0.002323	-0.239442	0.530595	-0.748281	0.659330
Subject5	-0.003752	-0.002168	0.002323	1.000000	-0.001059	0.007906	-0.000015	-0.003006
PC1	-0.760231	0.850822	-0.239442	-0.001059	1.000000	0.094441	-0.444118	-0.627490
PC2	-0.435569	0.118890	0.530595	0.007906	0.094441	1.000000	-0.406975	-0.230404
PC3	0.032495	-0.820452	-0.748281	-0.000015	-0.444118	-0.406975	1.000000	-0.279105
PC4	0.947770	-0.142587	0.659330	-0.003006	-0.627490	-0.230404	-0.279105	1.000000

Fig. A2 Correlation coefficients between each principal component and each speaker when PCA is applied to the July 22, 2021, meeting

	PC1
AI	0.186293
little bit	0.183401
that's right	0.182108
voice	0.181212
research	0.180585
Oh	0.178133
Speak	0.177453
Say	0.176751
called	0.176223
think	0.175723
information	0.174867
that	0.174867
channel	0.17458
CONFIDENTIAL	0.173501
individual	0.17277
maybe	0.170156
ICA	0.170006
Dr.Kimura	0.168764
just	0.167073
can	0.163604
human	0.161232

	PC2
already	0.232055
look	0.230957
some time ago	0.229324
correctly	0.225712
yes	0.224427
professor	0.223381
person	0.2233
there is	0.220558
write	0.2189
nice	0.206794
story	0.198506
various	0.196642
exist	0.184411
now	0.168289
voice	0.14629
ah	0.137249
I know	0.129184
understand	0.128924
tolerably	0.122155
just	0.121819
try	0.118857

	PC3
oh	0.17922
human	0.132568

Fig. A3 Words comprising each principal component (threshold at 0.1) when PCA is applied to the July 22, 2021, meeting

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